

- 4 (a) (i) By the principle of conservation,
gain in KE = loss in GPE

$$\frac{1}{2}mv^2 - 0 = mg\Delta h$$

$$\frac{1}{2}v^2 = (9.81)(0.30 - 0.30\cos 25^\circ)$$

$$v = 0.7426$$

$$v = 0.74 \text{ m s}^{-1}$$

$$\begin{aligned}
 \text{(ii)} \quad T - mg &= \frac{mv^2}{r} \\
 T &= \frac{(0.010)(0.74)^2}{0.30} + (0.010)(9.81) \\
 T &= 0.116 \text{ N}
 \end{aligned}$$

(iii) When the bob passes point B, the radius of the circular motion is smaller and the speed of the bob is the same.

Since the centripetal force is now larger, the tension is larger.

(b) (i) Vertically,

$$\begin{aligned}
 T \cos \theta &= mg \\
 (0.20) \cos \theta &= (0.010)(9.81) \\
 \theta &= 60.6^\circ
 \end{aligned}$$

(ii) Horizontally,

$$\begin{aligned}
 T \sin \theta &= m r \omega^2 \\
 (0.20) \sin \theta &= (0.010)(0.30 \sin \theta) \omega^2 \\
 \omega &= 8.16 \text{ rad s}^{-1}
 \end{aligned}$$